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Jeans-Independent Holdback Bypass for AMR Refinement in Cosmological Simulations

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ABSTRACT

Cosmological AMR simulations employ a *holdback* mechanism that progressively releases refinement levels as the universe expands, preventing premature over-refinement at high redshift. However, this mechanism blocks *all* refinement criteria simultaneously, including the Jeans length criterion, which is a numerical stability requirement rather than merely a resolution preference. When a holdback level is finally released, a large number of cells suddenly satisfy the refinement condition, triggering an abrupt starburst. We implement a `jeans_bypass_holdback` option that allows the Jeans criterion to operate independently of the holdback, with a configurable level limit (`n_jeans_bypass_levels`) to prevent uncontrolled refinement cascades. We discuss the interplay between the Jeans criterion, the polytropic equation of state, and the density threshold `d_jeans_thre`, showing that the polytropic temperature floor provides a natural self-regulation mechanism for the bypass.

Keywords: methods: numerical — cosmology: simulations — stars: formation — hydrodynamics

1. AMR REFINEMENT IN RAMSES

In RAMSES, each cell at level $\ell < N_{\rm levelmax}$ is tested for refinement at every coarse time-step. The decision to subdivide a cell into $2^{N_{\rm dim}}$ children is governed by a combination of physical, geometric, and cosmological criteria.

1.1. *Refinement Logic*

The overall refinement flag for a cell is determined by the following Boolean expression:

$$\mathcal{R} = \neg\,\mathcal{H} \;\wedge\; \mathcal{G} \;\wedge\; \big(\mathcal{M} \vee \mathcal{S} \vee \mathcal{D} \vee \mathcal{P} \vee \mathcal{U} \vee \mathcal{J}\big), \tag{1}$$

where each symbol is defined below.

1.1.1. *Physical Criteria (OR)*

1. **Quasi-Lagrangian mass criterion ($\mathcal{M}$).** The cell is flagged if its gas density exceeds a level-dependent threshold:

$$\rho_{\rm cell} \;\geq\; m_{\rm refine}(\ell)\,\frac{m_{\rm sph}}{V_{\rm cell}}, \tag{2}$$

   where $m_{\rm sph}$ is the effective SPH kernel mass and $V_{\rm cell} = (\Delta x_\ell)^3$. This criterion ensures that regions containing roughly $m_{\rm refine}$ or more particles per cell are refined, providing a quasi-Lagrangian resolution that tracks gravitational collapse. Controlled by the namelist array `m_refine(1:nlevelmax)`.

2. **Sink particle forcing ($\mathcal{S}$).** If `sink_refine=.true.`, any cell containing a sink cloud particle (identified by `idp` < 0 and `tp` $= 0$) is unconditionally flagged up to $N_{\rm levelmax}$, guaranteeing maximum resolution around each accretion region.

3. **Density gradient ($\mathcal{D}$).** The normalised density gradient between a cell and each of its $2N_{\rm dim}$ face-neighbours is computed as

$$\epsilon_\rho = 2\,\max\!\left(\frac{|\rho_+ - \rho_0|}{|\rho_+ + \rho_0| + \rho_{\rm floor}},\; \frac{|\rho_0 - \rho_-|}{|\rho_0 + \rho_-| + \rho_{\rm floor}}\right). \tag{3}$$

   If $\epsilon_\rho >$ `err_grad_d` along any axis, the cell is flagged.

4. **Pressure gradient ($\mathcal{P}$).** Analogous to the density gradient but applied to the thermal pressure:

$$\epsilon_P = 2\,\max\left(\frac{|P_+ - P_0|}{|P_+ + P_0| + P_{\rm floor}},\ \frac{|P_0 - P_-|}{|P_0 + P_-| + P_{\rm floor}}\right), \tag{4}$$

flagging when $\epsilon_P >$ `err_grad_p`.

5. **Velocity gradient ($\mathcal{U}$).** A normalised velocity jump relative to the local sound speed:

$$\epsilon_v = 2\,\max\left(\frac{|v_+ - v_0|}{c_{s,+} + c_{s,0} + |v_+| + |v_0| + v_{\rm floor}},\ \frac{|v_0 - v_-|}{c_{s,0} + c_{s,-} + |v_0| + |v_-| + v_{\rm floor}}\right), \tag{5}$$

flagging when $\epsilon_v >$ `err_grad_u` in any component.

6. **Jeans length ($\mathcal{J}$).** When `jeans_refine`$(\ell) = N_J > 0$, the cell is flagged if the local Jeans length is shorter than N_J cell widths:

$$\lambda_{\rm J} = \sqrt{\frac{\pi\, c_s^2}{G\,\rho}} < N_J\,\Delta x_\ell, \tag{6}$$

implementing the Truelove et al. (1997) condition to prevent artificial fragmentation in self-gravitating gas. An additional density threshold `d_jeans_thre` restricts this criterion to cells with $n_{\rm H} \geq$ `d_jeans_thre`.

1.1.2. *Geometry Mask (AND)*

Ellipsoidal region ($\mathcal{G}$). When `r_refine`$(\ell) > -1$, only cells inside a generalised ellipsoidal region are permitted to refine. The test is

$$\left(\frac{2|\Delta x|}{r}\right)^{e_r} + \left(\frac{2|\Delta y|}{a\,r}\right)^{e_r} + \left(\frac{2|\Delta z|}{b\,r}\right)^{e_r} < 1, \tag{7}$$

where $r =$ `r_refine`, $a =$ `a_refine`, $b =$ `b_refine`, and $e_r =$ `exp_refine`. Cells outside this region are *un-flagged* regardless of physical criteria, restricting high resolution to a zoom-in target.

1.1.3. *Cosmological Holdback (NOT)*

Resolution holdback ($\mathcal{H}$). In cosmological runs with cooling, overly eager refinement at high redshift can exhaust memory before the epoch of interest. When `q_refine_holdback=.true.` and the physical cell size $\Delta x_\ell < 4^{1/N_{\rm dim}}\,\Delta x_{N_{\rm levelmax}}/a_{\rm exp}$ for levels $\ell >$ `nlevelmax_part` $+\,3$, the *entire* level is blocked from further refinement:

$$\mathcal{H}(\ell) = \big[\ell > \ell_{\rm part} + \ell_{\rm collapse}\big]\ \wedge\ \big[\Delta x_\ell < 4^{1/N_{\rm dim}}\,\Delta x_{\rm min}/a\big], \tag{8}$$

where $\ell_{\rm part}$ is the particle initial-condition level (typically $\ell_{\rm min}$) and $\ell_{\rm collapse} = 3$. The first holdback level $\ell_{\rm hb}$ is the smallest ℓ satisfying Equation (8):

$$\ell_{\rm hb} = \min\big\{\ell > \ell_{\rm part} + \ell_{\rm collapse}\ :\ \Delta x_\ell < 4^{1/N_{\rm dim}}\,\Delta x_{\rm min}/a\big\}. \tag{9}$$

1.2. *Buffer Zones*

After applying Equation (1), flagged cells are expanded by `nexpand`(ℓ) cell widths in each direction via a dilation operator (`smooth_fine`). This cubic buffer ensures smooth refinement boundaries, preventing the Godunov solver from encountering abrupt level transitions.

2. THE STARBURST PROBLEM

The holdback mechanism blocks *all* refinement criteria at levels $\ell \geq \ell_{\rm hb}$, including the Jeans criterion. This creates two related problems:

Artificial fragmentation.—The Jeans length criterion $\mathcal{J}$ differs fundamentally from the other refinement triggers: it is a *numerical stability* requirement (Truelove et al. 1997), not merely a resolution preference. Violating the Jeans condition leads to artificial fragmentation — a qualitatively wrong result — regardless of epoch. Yet in the standard RAMSES logic (Equation 1), the cosmological holdback $\mathcal{H}$ blocks $\mathcal{J}$ uniformly with all other criteria. At high redshift, dense self-gravitating regions may therefore be held at insufficient resolution, producing spurious fragmentation that propagates into lower-redshift structure.

Table 1. Cascade refinement without level limit ($a = 0.164$, $\ell_{\rm hb} = 14$, $\ell_{\rm max} = 17$). All levels ≥ 14 are in the holdback zone.

Level	$\Delta x/\Delta x_{\rm min}$	Holdback	Bypass	$\mathcal{J}$ fires	Result
≤ 12	≥ 32	No	—	Yes (normal)	Normal
13	16	No	—	Yes (normal)	Normal
14	8	Yes	Active	Yes	→ lv. 15 created
15	4	Yes	Active	Yes	→ lv. 16 created
16	2	Yes	Active	Yes	→ lv. 17 created
17	1	$\ell_{\rm max}$	—	—	OOM crash

Abrupt SFR transition.—When the holdback releases level ℓ (i.e. $\mathcal{H}(\ell)$ transitions from true to false), all accumulated criteria fire simultaneously. Many cells that have been gravitationally collapsing during the holdback period now suddenly satisfy the refinement condition, leading to a burst of new cells and an abrupt spike in the star formation rate.

3. JEANS BYPASS: DESIGN AND IMPLEMENTATION

3.1. *Modified Refinement Logic*

The key observation is that the Jeans length criterion is *cell-local*: it depends only on the density and temperature of the cell itself (`uold`), with no neighbor information required. This allows a lightweight evaluation even when the full refinement machinery (which requires neighbor gathering for $\mathcal{D}$, $\mathcal{P}$, $\mathcal{U}$) is skipped by the holdback.

With `jeans_bypass_holdback=.true.`, the refinement logic becomes

$$\mathcal{R}' = \Big[\neg\mathcal{H} \;\wedge\; \mathcal{G} \;\wedge\; \big(\mathcal{M} \vee \mathcal{S} \vee \mathcal{D} \vee \mathcal{P} \vee \mathcal{U} \vee \mathcal{J}\big)\Big] \;\vee\; \Big[\mathcal{H} \;\wedge\; \mathcal{J} \;\wedge\; (\ell < \ell_{\rm hb} + n_{\rm bypass})\Big], \tag{10}$$

where $n_{\rm bypass} \equiv$ `n_jeans_bypass_levels` (default 1). The geometry mask $\mathcal{G}$ still applies through the non-holdback branch, preserving zoom-in spatial restrictions. Only the temporal holdback $\mathcal{H}$ is bypassed for $\mathcal{J}$, and only within a limited number of levels.

This modification is analogous to the existing treatment of sink particles: $\mathcal{S}$ is already evaluated inside the holdback-protected block, but unlike $\mathcal{J}$ it only triggers in cells that already host a sink. Extending the same exemption to $\mathcal{J}$ ensures numerical integrity for gas that is *about to* form a sink.

3.2. *Level Limitation: Preventing Cascades*

Without a level limit ($n_{\rm bypass} \to \infty$), the Jeans bypass creates a catastrophic cascade. When a cell at the holdback level $\ell_{\rm hb}$ is flagged by the Jeans criterion, child cells are created at level $\ell_{\rm hb} + 1$. Since level $\ell_{\rm hb} + 1$ is also in the holdback zone, the bypass applies again, flagging cells at $\ell_{\rm hb} + 2$, and so on until $\ell_{\rm max}$. This rapidly exhausts available memory.

The cascade occurs because the Jeans length in radiatively cooling gas (below the star-formation density threshold) shrinks without bound as the temperature drops. There is no physical mechanism to halt the Jeans criterion from triggering at every successive level — only the polytropic temperature floor (Section 4) or a hard level limit can stop it.

Table 1 shows the progression observed in a test run without level limit.

The memory footprint grew from 81% to 88.5% in a single coarse step before crashing. The fine-step timestep also decreased by 8× due to subcycling at levels 15–17.

With $n_{\rm bypass} = 1$, only level $\ell_{\rm hb} = 14$ is bypassed, creating cells at level 15 but no further. This provides a controlled pre-refinement before the holdback releases.

Table 2. Two complementary mechanisms for preventing the bypass cascade.

Mechanism	Type	Behaviour
`n_jeans_bypass_levels`	Hard limit	Caps extra levels regardless of physics
`d_jeans_thre` $\approx n_\star$	Physics-based	Polytropic floor limits $\lambda_{\rm J}$ decrease

4. ROLE OF THE POLYTROPIC TEMPERATURE FLOOR

4.1. *Jeans Length in the Polytropic Regime*

In RAMSES, the star-formation module enforces a polytropic equation of state above the density threshold $n_\star$:

$$T(n) = T_{2,\star}\left(\frac{n}{n_\star}\right)^{g_\star - 1}, \qquad n \geq n_\star, \tag{11}$$

with default parameters $T_{2,\star} = 750\,\mathrm{K}$, $n_\star = 0.1\,\mathrm{cm}^{-3}$, $g_\star = 5/3$. This floor is applied in the cooling routine and is already reflected in `uold`, so the Jeans length computation in `jeans_length_refine()` automatically includes the polytropic stabilisation.

Substituting Equation (11) into Equation (6):

$$\lambda_{\rm J} \propto \sqrt{\frac{T}{\rho}} \propto \sqrt{\rho^{g_\star - 1}/\rho} = \rho^{(g_\star - 2)/2}. \tag{12}$$

For $g_\star = 5/3$: $\lambda_{\rm J} \propto \rho^{-1/6}$. The Jeans length *decreases very slowly* with density — one or two additional levels of refinement are typically sufficient to resolve it.

In contrast, for gas below $n_\star$ (no polytropic floor), radiative cooling can drive T to very low values while ρ remains moderate. This yields $\lambda_{\rm J} \ll \Delta x_\ell$ at many levels, causing the cascade.

4.2. *Self-regulation via* `d_jeans_thre`

The density threshold parameter `d_jeans_thre` (in units of $n_{\rm H}\,[\mathrm{cm}^{-3}]$) restricts the Jeans criterion to cells above a given density:

$$\mathcal{J}(\text{cell}) = \begin{cases} N_J\,\Delta x_\ell \geq \lambda_{\rm J} & \text{if } n \geq \texttt{d_jeans_thre}, \\ \text{false} & \text{otherwise.} \end{cases} \tag{13}$$

Setting `d_jeans_thre` $\approx n_\star$ restricts the Jeans criterion to the polytropic regime. The cascade then self-limits through the following mechanism:

1. At level $\ell_{\rm hb}$: only cells with $n \geq n_\star$ are checked. The polytropic floor guarantees $T \geq T_{2,\star}$, so $\lambda_{\rm J}$ is relatively large.
2. If $N_J\,\Delta x_{\ell_{\rm hb}} > \lambda_{\rm J}$, the cell is flagged and children at $\ell_{\rm hb} + 1$ are created.
3. At level $\ell_{\rm hb} + 1$: Δx is halved, density is approximately unchanged, $\lambda_{\rm J}$ is approximately unchanged.
4. $N_J\,\Delta x_{\ell_{\rm hb}+1} = N_J\,\Delta x_{\ell_{\rm hb}}/2 < \lambda_{\rm J} \Rightarrow$ Jeans criterion is *satisfied* $\Rightarrow$ no further flagging $\Rightarrow$ cascade stops naturally.

This provides a *physics-based* self-regulation, complementary to the *hard limit* imposed by `n_jeans_bypass_levels`.

When `d_jeans_thre` $= 0$ (default), *all* cells are checked, including low-density gas where cooling produces arbitrarily small $\lambda_{\rm J}$. This is the root cause of the unlimited cascade observed in Section 3.2.

5. IMPLEMENTATION DETAILS

5.1. *Code Structure*

The implementation modifies two source files (plus parameter declaration and namelist reading).

flag_utils.kjhan.f90 — *userflag_fine()*—The holdback early-return is modified from a simple `return` to a conditional block:

```
if(prevent_refine .and. q_refine_holdback) then
   if(jeans_bypass_holdback .and. hydro &
      .and. n_jeans_bypass_levels > 0) then
      ! Find the first holdback level
      dx_min = (0.5D0**nlevelmax)*scale
      holdback_threshold = &
        (4d0**(1d0/ndim))*(dx_min/aexp)
      holdback_first = nlevelmax + 1
      do jlevel = nlevelmax_part &
         + nlevel_collapse+1, nlevelmax
         dx_test = (0.5d0**jlevel)*scale
         if(dx_test < holdback_threshold) then
            holdback_first = jlevel
            exit
         endif
      end do
      ! Only bypass within allowed range
      if(ilevel < holdback_first &
         + n_jeans_bypass_levels) then
         call jeans_only_flag(ilevel)
      endif
   endif
   return
endif
```

hydro_flag.kjhan.f90 — *jeans_only_flag()*—A new lightweight subroutine that:

1. Loops over active grids with OpenMP parallelism.
2. Computes cell indices without neighbor gathering.
3. Calls the existing `jeans_length_refine()` subroutine (no code duplication).
4. Sets `flag1` for flagged cells and updates `nflag`.

This is possible because `jeans_length_refine()` only accesses cell-local data (`uold(ind_cell,...)`), unlike the gradient-based criteria $\mathcal{D}$, $\mathcal{P}$, $\mathcal{U}$.

5.2. *Safety Properties*

- **Default off**: With `jeans_bypass_holdback=.false.` (default), the new code path is never entered. Bit-identical to the unmodified code.
- **No neighbor access**: `jeans_only_flag()` is safe to call at any point in the refinement cycle.
- **OMP safety**: `flag1` writes are grid-partitioned; `nflag` uses reduction.
- **Hard level limit**: `n_jeans_bypass_levels` provides a fail-safe even if `d_jeans_thre` is misconfigured.

6. SUMMARY OF PARAMETERS

Table 3 lists all refinement-related namelist parameters, including the new bypass parameters.

The recommended configuration for cosmological runs with star formation is:

```
&REFINE_PARAMS
m_refine          = 9*8.,
jeans_refine      = 9*4.,
jeans_bypass_holdback  = .true.
n_jeans_bypass_levels  = 1
d_jeans_thre      = 0.1     ! = n_star
q_refine_holdback = .true.
/
```

7. TEST RESULTS

All tests use the HR5 medium-box setup: $\ell_{\min} = 9$, $\ell_{\max} = 17$, 12 MPI ranks, OMP=5, restarting from output 5 ($a = 0.164$, $z \approx 5.1$, nstep_coarse=241).

Table 3. AMR refinement parameters in the `&AMR_PARAMS` and `&REFINE_PARAMS` namelists. New parameters are marked with ⋆.

Parameter	Default	Role
`m_refine`	8.0	Lagrangian mass threshold (per level)
`sink_refine`	F	Force max level around sinks
`err_grad_d`	−1	Density gradient threshold
`err_grad_p`	−1	Pressure gradient threshold
`err_grad_u`	−1	Velocity gradient threshold
`jeans_refine`	−1	Jeans length in cells (per level)
`d_jeans_thre`	0	Density threshold for Jeans ($n_{\rm H}\,{\rm cm}^{-3}$)
`r_refine`	−1	Geometry region diameter (per level)
`a_refine`	1.0	Ellipticity Y/X
`b_refine`	1.0	Ellipticity Z/X
`exp_refine`	2.0	Norm exponent
`x/y/z_refine`	0.5	Region centre
`nexpand`	1	Buffer zone width (per level)
`q_refine_holdback`	F	Cosmological holdback
⋆ `jeans_bypass_holdback`	F	Enable Jeans bypass of holdback
⋆ `n_jeans_bypass_levels`	1	Max extra levels beyond $\ell_{\rm hb}$

7.1. *Bit-identical Verification*

With `jeans_bypass_holdback=.false.` (default) and no `jeans_refine`, the new binary produces results identical to the reference run: econs=6.13×10^{-4} at step 242, econs=6.71×10^{-4} at step 243.

7.2. *Unlimited Bypass*

With `jeans_refine=9*4.`, `jeans_bypass_holdback=.true.`, `d_jeans_thre=0`, and no level limit: cascade refinement from level 14 through 17 within a single coarse step. Memory grew from 81% to 88.5% before OOM crash. Fine-step timestep decreased by 8× due to subcycling.

7.3. *Limited Bypass ($n_{\rm bypass} = 1$)*

With `n_jeans_bypass_levels=1`: only level 14 is bypassed, cells at level 15 may be created but no further cascade.

7.4. *Future: `d_jeans_thre` $= n_\star$*

A test with `d_jeans_thre=0.1` will verify the physics-based self-regulation, potentially allowing $n_{\rm bypass} > 1$ without cascade, since only gas in the polytropic regime would trigger the Jeans criterion.

8. COOLING MODULE: LIMITATIONS AND ALTERNATIVES

The cooling module is a critical component of the simulation. It determines the equilibrium temperature of the gas, which in turn controls the Jeans length $\lambda_{\rm J} \propto \sqrt{T/\rho}$ and hence the refinement decisions through the Jeans criterion $\mathcal{J}$. Errors in the cooling calculation — whether from the cooling rate tables, the UV background model, or the numerical integration scheme — propagate directly into errors in $\lambda_{\rm J}$ and may trigger (or suppress) refinement at the wrong density and temperature.

In this section we first analyse the limitations of the standard RAMSES cooling solver, then compare its underlying cooling rates against the Grackle equilibrium chemistry library (Smith et al. 2017), and finally describe an alternative exact integration scheme based on the Townsend (2009) Y-function method (Townsend 2009).

8.1. *The Standard* RAMSES *Cooling Solver*

The RAMSES cooling module solves the energy equation

$$\frac{dT_2}{dt} = -\frac{2X_{\rm H}}{3k_{\rm B}}\,\frac{\Lambda_{\rm net}(n_{\rm H}, T_2)}{n_{\rm H}}, \tag{14}$$

where $T_2 \equiv T/\mu$ is the temperature per mean molecular weight and $\Lambda_{\rm net}$ is the net volumetric cooling rate (cooling − heating) in $\mathrm{erg\,cm^{-3}\,s^{-1}}$.

The solver uses a semi-implicit Newton–Raphson iteration. At each sub-step, the temperature is updated as

$$T_2^{(n+1)} = T_2^{(n)}\,\frac{1 + \Lambda'/w_{\rm cool} - \Lambda/(T_2^{(n)}\,w_{\rm cool})}{1 + \Lambda'/w_{\rm cool}}, \tag{15}$$

where $\Lambda' \equiv d\Lambda_{\rm net}/dT_2$ and the adaptive step control parameter is $w_{\rm cool} = \max\big(|\Lambda|/(T_2\,\Delta_{\rm var}),\ w_{\rm max},\ -\Lambda'\,\Delta_{\rm var}\big)$ with $\Delta_{\rm var} = 4$.

The cooling rates $\Lambda_{\rm net}$ and their derivatives are interpolated from a pre-computed 2D table $\Lambda(n_{\rm H}, T_2)$, built at the start of the simulation for the current expansion factor a. The table contains separate contributions from:

- H/He collisional cooling (ionisation, recombination, bremsstrahlung, excitation) computed from collisional ionisation equilibrium (CIE);
- UV photo-heating from the Courty & Alimi (2004) background model;
- Compton cooling/heating off the CMB;
- metal cooling from the Cloudy 07 table (Plöckinger & Schaye 2015), scaled linearly with metallicity Z.

8.1.1. *Overshooting Near Thermal Equilibrium*

The Newton–Raphson scheme (Equation 15) is designed for monotonic cooling (or heating). However, near the thermal equilibrium temperature $T_{\rm eq}$ where $\Lambda_{\rm net}(T_{\rm eq}) = 0$, the net rate changes sign and the update can overshoot. Specifically:

Sign reversal.—Consider a cell at temperature $T > T_{\rm eq}$ that is cooling ($\Lambda_{\rm net} > 0$). A Newton–Raphson step decreases T_2. If the step is large enough to cross $T_{\rm eq}$, the cell enters the heating regime ($\Lambda_{\rm net} < 0$). On the next iteration, the solver increases T_2, potentially overshooting back above $T_{\rm eq}$. The result is an oscillation around $T_{\rm eq}$ that converges only after many sub-steps, or in the worst case does not converge within the 500-iteration limit.

Steep cooling function.—The problem is exacerbated where the cooling function has a steep slope — in particular, near the hydrogen Lyman-α peak at $T \sim 10^4\,\mathrm{K}$ and the metal cooling peak at $T \sim 10^{4.5}$–$10^{5.5}\,\mathrm{K}$. In these temperature ranges, $|\Lambda'|$ is large and the linearised N-R step can dramatically overshoot. The adaptive step control $w_{\rm cool}$ mitigates this by limiting $\Delta T_2/T_2 \le \Delta_{\rm var}$, but for cells undergoing rapid cooling (large $|\Lambda|/T_2$), the effective sub-step can still be too large to track the equilibrium accurately.

Cumulative error.—Because T_2 enters the next hydrodynamic time-step as the updated thermal energy, any overshoot at step n feeds back into the pressure and hence the Jeans length at step n+1. A cell that overshoots to $T < T_{\rm eq}$ will have an artificially low Jeans length, potentially triggering spurious refinement. Conversely, a cell that undershoots (remains at $T > T_{\rm eq}$) may fail to trigger the Jeans criterion when it should.

Time-step coupling.—The maximum sub-step size within `solve_cooling` is determined by the cooling time $t_{\rm cool} = (3/2)\,nk_{\rm B}T/|\Lambda_{\rm net}|$. When $\Lambda_{\rm net}$ is large, the solver takes many small sub-steps and is accurate. But near equilibrium, $|\Lambda_{\rm net}|$ is small, $t_{\rm cool}$ is large, and the solver takes fewer, larger steps — precisely where accuracy matters most for determining the equilibrium temperature.

8.1.2. *Limitations of the 2D Table*

Beyond the numerical integration, the cooling *rates* themselves have systematic biases:

1. **No metallicity axis.** The RAMSES table is 2D: $(n_{\rm H}, T_2)$. Metal cooling is stored as a separate 1D function $\Lambda_Z(T)$ at solar metallicity, then scaled as $Z \cdot \Lambda_Z$. This linear scaling is correct for CIE but neglects the density-dependent ionisation corrections that become important at $n_{\rm H} > 1\,{\rm cm}^{-3}$, where photo-ionisation by the UV background cannot penetrate and the gas is self-shielded.

2. **CIE assumption.** The mean molecular weight μ is computed assuming collisional ionisation equilibrium at each $(n_{\rm H}, T)$ point. This neglects time-dependent ionisation (important during rapid cooling) and photo-ionisation at low temperatures (important in the diffuse IGM at $T < 10^4\,{\rm K}$), where the UV background maintains a residual ionisation fraction that CIE would predict to be negligible.

3. **UV background model.** The default Courty model is a polynomial fit calibrated against Cloudy 07. More recent UV background models (e.g. Haardt & Madau 2012; Faucher-Giguère 2020) differ significantly at $z > 3$, particularly in the He II reionisation epoch ($z \sim 3$–4) and at the earliest epochs where the UV intensity rises steeply.

4. **Temperature variable $T_2 = T/\mu$.** The solver works with T_2 rather than T. Since μ changes across the ionisation transition ($\mu \approx 1.22$ for neutral gas, $\mu \approx 0.59$ for fully ionised), a small change in T_2 maps to a factor-of-two change in physical temperature T across the transition. This amplifies interpolation errors near $T \sim 10^4\,{\rm K}$.

8.2. *Comparison on the Density–Temperature Plane*

To quantify the systematic differences, we compare the RAMSES net cooling rate against the Grackle equilibrium cooling library (Smith et al. 2017), which uses the Haardt & Madau (2012) UV background (Haardt & Madau 2012) and includes density-dependent metal cooling with full non-LTE ionisation corrections. We compute $\Lambda_{\rm net}(n_{\rm H}, T)$ on a grid of 60×256 points spanning $n_{\rm H} = 10^{-8}$–$10^4\,{\rm cm}^{-3}$ and $T = 10$–$10^9\,{\rm K}$ at $z = 5.1$ (the redshift of the HR5 test restart).

Low metallicity ($Z = 10^{-3}\,Z_\odot$).—Figure 1 compares the two models at the simulation average metallicity (`z_ave` = 10^{-3}). At this low metallicity, the net rate is dominated by H/He processes and UV photo-heating.

The lower-left panel shows the log ratio of the cooling rates, $\log_{10}(|\Lambda_{\rm RAMSES}|/|\Lambda_{\rm Grackle}|)$. Contour lines mark the thermal equilibrium ($\Lambda_{\rm net} = 0$) for each model: the yellow solid line for RAMSES and the green dashed line for Grackle. Three features are apparent:

1. The *equilibrium temperature* differs by up to $\Delta \log T \sim 0.3$–0.5 at low densities ($n_{\rm H} \lesssim 10^{-2}\,{\rm cm}^{-3}$), reflecting the different UV backgrounds. Since $\lambda_{\rm J} \propto T^{1/2}$, this translates to a ~ 40–80% difference in the Jeans length at the same density.

2. At intermediate temperatures (10^4–$10^5\,{\rm K}$), the RAMSES rate is systematically *stronger* (redder in the ratio panel) by up to an order of magnitude. This arises because the CIE assumption overpredicts the neutral hydrogen fraction at these temperatures for low-density gas where the UV background maintains significant photo-ionisation.

3. At $T > 10^7\,{\rm K}$, both models converge to the bremsstrahlung limit $\Lambda \propto T^{1/2}\,n_{\rm H}^2$ and the ratio approaches unity, as expected.

The lower-right panel shows the mean molecular weight difference $\Delta\mu = \mu_{\rm RAMSES} - \mu_{\rm Grackle}$. The RAMSES CIE μ is systematically higher than the Grackle non-LTE value at $T > 10^5\,{\rm K}$. At intermediate temperatures ($T \sim 10^4$–$10^5\,{\rm K}$), the Grackle μ shows a strong *density dependence* due to the UV background maintaining partial ionisation at low densities — a feature entirely absent from the RAMSES model, where $\mu(T_2)$ is independent of density. This density-dependent μ has direct implications for the $T_2 \to T$ conversion and hence for the Jeans length evaluation.

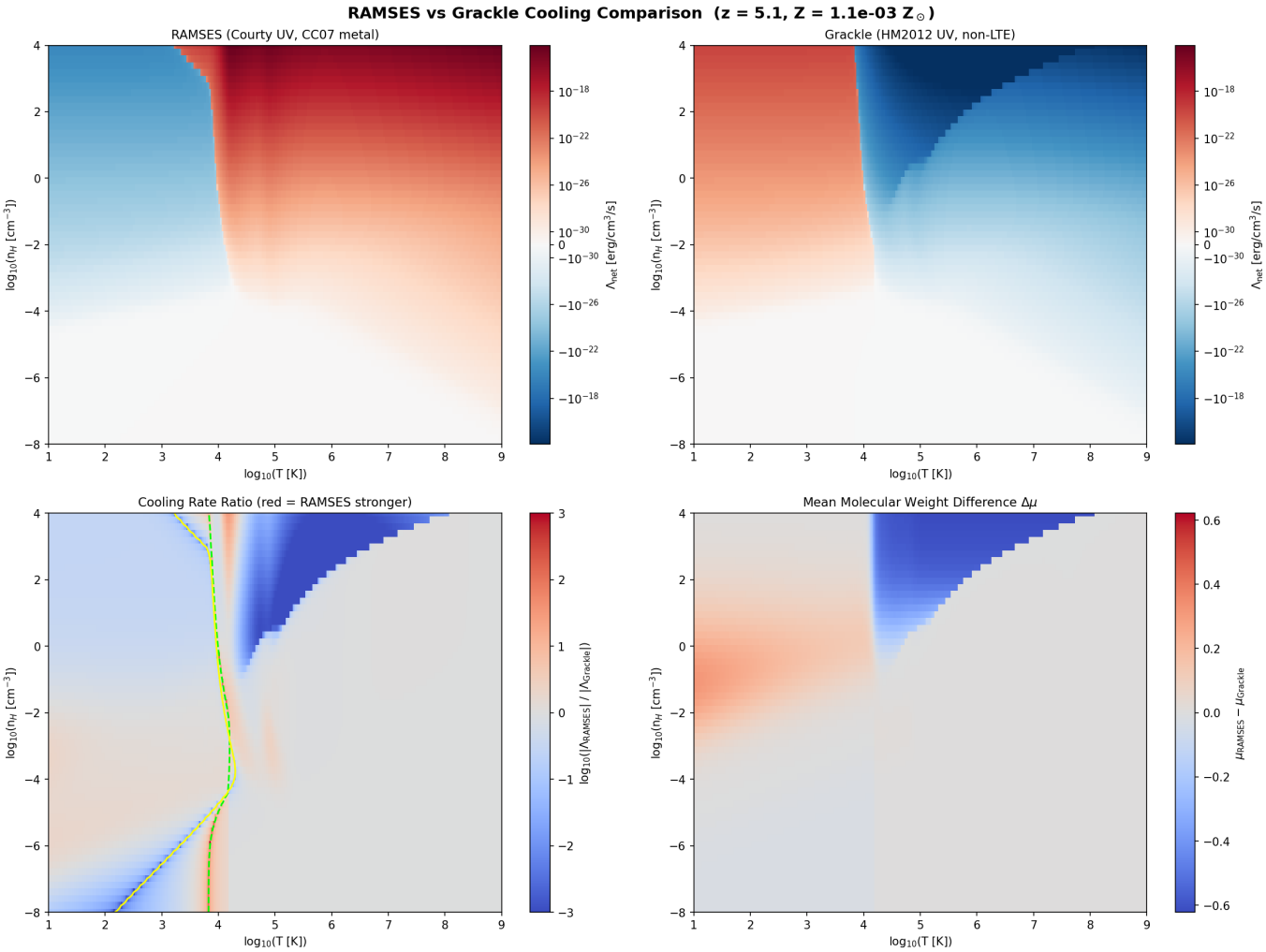

Figure 1. Comparison of the RAMSES and Grackle net cooling rates at $z = 5.1$ and $Z = 1.1 \times 10^{-3}\, Z_\odot$ (the simulation mean metallicity). *Top left:* RAMSES net cooling rate (Courty UV, Cloudy 07 CIE metals). *Top right:* Grackle net cooling rate (HM2012 UV, non-LTE equilibrium chemistry). Red shading indicates net cooling (energy loss); blue indicates net heating (energy gain). *Bottom left:* Log ratio of the absolute cooling rates, $\log_{10}(|\Lambda_{\rm RAMSES}|/|\Lambda_{\rm Grackle}|)$; yellow solid curve = RAMSES thermal equilibrium ($\Lambda = 0$), green dashed = Grackle equilibrium. Note the divergence of the equilibrium contours at $n_{\rm H} < 1\,{\rm cm}^{-3}$. *Bottom right:* Mean molecular weight difference $\mu_{\rm RAMSES} - \mu_{\rm Grackle}$, showing the density-dependent μ from Grackle's non-LTE treatment.

Solar metallicity ($Z \approx Z_\odot$).—Figure 2 shows the same comparison at near-solar metallicity, where metal line cooling dominates the temperature range $T \sim 10^4$–$10^{5.5}$ K. The differences between the two models are significantly amplified:

- The Grackle table includes density-dependent ionisation corrections for metal species, leading to *up to* $\sim 10\times$ *stronger cooling* in the $T \sim 10^{4.5}$–10^5 K range compared to the CC07 CIE table. This is precisely the temperature range where infalling gas cools through the virial temperature of 10^{11}–$10^{12}\, M_\odot$ haloes.

- The equilibrium temperature contours diverge more strongly, with the RAMSES equilibrium systematically higher at $n_{\rm H} < 10^{-1}\,{\rm cm}^{-3}$. This implies that the standard RAMSES solver overestimates the temperature of the warm-hot intergalactic medium (WHIM), reducing the Jeans-based refinement trigger and potentially missing structure in filaments.

- The UV suppression factor for metal cooling differs: the Courty model applies a redshift-dependent attenuation $f_{\rm UV}(n_{\rm H}, z)$ that damps metal cooling more strongly at low densities than the Grackle self-shielding prescription.

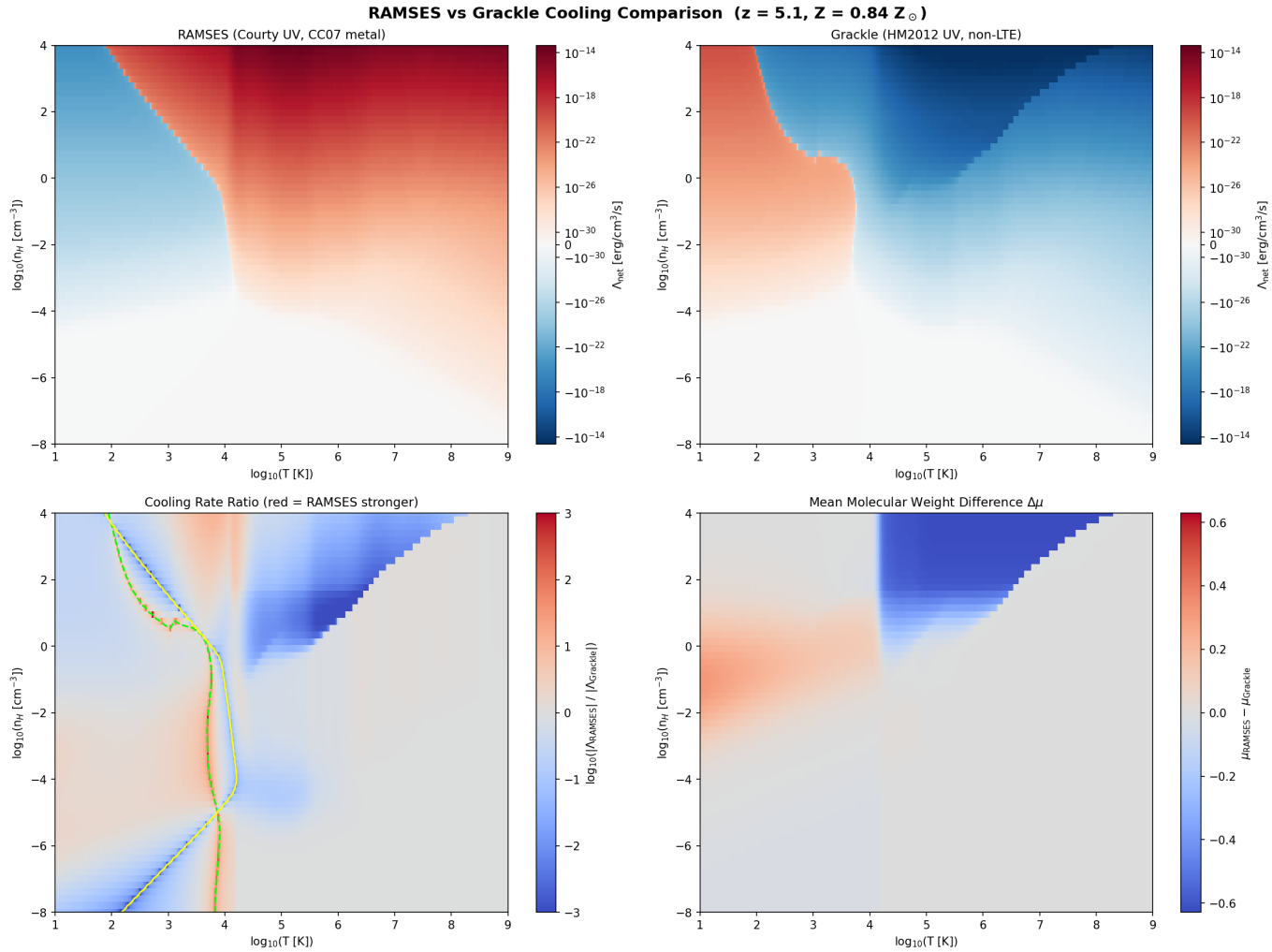

Figure 2. Same as Figure 1 but at solar metallicity ($Z = 0.84\,Z_\odot$). Metal cooling dominates the net rate at $T \sim 10^4$–$10^{5.5}$ K, amplifying the model differences. Note the larger separation of the equilibrium contours (bottom-left) and the stronger $\Delta\mu$ pattern (bottom-right) compared to the low-metallicity case.

8.3. *The Townsend Exact Integration Scheme*

To address the overshooting problem of the N-R solver (Section 8.1.1), we implement the exact integration method of Townsend (2009), which transforms the cooling ODE into an analytically invertible form using the Y-function.

8.3.1. *Y-Function Formulation*

For a piecewise power-law cooling function $\Lambda(T) \propto T^{\alpha_j}$ in each temperature bin $[T_j, T_{j+1}]$, the integrated cooling variable is

$$Y(T) = \int_{T_{\rm ref}}^{T} \frac{dT'}{T'\,\hat{\Lambda}(T')} = \sum_k \Delta Y_k, \tag{16}$$

where $\hat{\Lambda} = \Lambda/\Lambda_{\rm ref}$ is the normalised cooling rate. Each segment contributes

$$\Delta Y_j = \begin{cases} \dfrac{T_j^{-\alpha_j} - T_{j+1}^{-\alpha_j}}{\alpha_j\,\hat{\Lambda}_j\,T_j^{-\alpha_j}} & (\alpha_j \neq 0), \\ \dfrac{\ln(T_j/T_{j+1})}{\hat{\Lambda}_j} & (\alpha_j = 0). \end{cases} \tag{17}$$

The exact temperature after a time-step Δt is obtained by inverting the relation

$$Y(T^{n+1}) = Y(T^n) + \Lambda_{\rm ref}\,\frac{n_{\rm H}}{E_0}\,\Delta t, \tag{18}$$

where $E_0 = (3/2)\,n_{\rm tot}\,k_{\rm B}$ is the thermal energy per unit temperature. The inversion is performed by binary search on the cumulative Y-table and local analytic inversion within the identified segment.

8.3.2. *Advantages Over Newton–Raphson*

The Y-function scheme has several fundamental advantages:

1. **No overshooting.** The method integrates the cooling equation *exactly* for piecewise power-law rates. The temperature evolves monotonically toward the equilibrium; it cannot oscillate across $T_{\rm eq}$.
2. **No sub-stepping.** A single evaluation of Equation (18) advances the solution by the full hydrodynamic time-step, regardless of $t_{\rm cool}/\Delta t$. The N-R solver requires up to 500 sub-steps when $\Delta t \gg t_{\rm cool}$.
3. **Correct equilibrium.** When $\Delta t \to \infty$, $Y \to Y_{\rm eq}$ and the cell relaxes exactly to $T_{\rm eq}$. The N-R solver only reaches $T_{\rm eq}$ asymptotically through iteration, and may settle on the wrong side if the sub-step structure interacts with the sign change.
4. **Deterministic.** The result depends only on $(T^n, n_{\rm H}, Z, \Delta t)$ and the cooling table, with no sensitivity to iteration count, convergence tolerance, or adaptive step heuristics.

The cost is a table lookup and a log / exp evaluation per cell (∼few μs), comparable to a single N-R sub-step. Since the N-R solver typically requires 5–50 sub-steps, the exact scheme is faster for strongly cooling gas.

8.3.3. *Implementation: Eunha Cooling Module*

The exact integrator is implemented as a Fortran module (`eunha_cooling_mod.f90`) that reads a 3D Grackle equilibrium cooling table $\Lambda_{\rm net}(n_{\rm H}, Z, T)$ and pre-computes the cumulative Y-table. The module provides three runtime modes selected by the namelist parameter `cooling_method`:

- `'original'` — standard RAMSES N-R solver (default, no overhead);
- `'exact'` — Townsend exact integration replaces the N-R solver;
- `'compare'` — both solvers run; the N-R result is used for physics while per-step diagnostics report the difference $\Delta T_2^{\rm orig} - \Delta T_2^{\rm exact}$.

The Grackle table adds a third axis (metallicity Z) with 40 bins spanning 10^{-6}–$10\,Z_\odot$, compared to the 2D RAMSES table with linear Z scaling. The mean molecular weight $\mu(n_{\rm H}, Z, T)$ is also tabulated from Grackle's non-LTE equilibrium, replacing the RAMSES CIE calculation. The $T_2 \leftrightarrow T$ conversion uses an iterative scheme: $\mu \to T = T_2\,\mu \to \mu'(n_{\rm H}, Z, T) \to T' = T_2\,\mu'$, converging to $< 0.1\%$ in 3 iterations.

8.4. *Implications for the Jeans Criterion*

The combined effect of the cooling rate differences (Figures 1–2) and the integration scheme impacts the Jeans refinement criterion through three channels:

1. Equilibrium temperature.—The Jeans length at thermal equilibrium is $\lambda_{\rm J}^{\rm eq} \propto (T_{\rm eq}/\rho)^{1/2}$. Since $T_{\rm eq}^{\rm RAMSES} \neq T_{\rm eq}^{\rm Grackle}$ (bottom-left panels), the two solvers will flag different cells. At low densities, the RAMSES equilibrium is systematically *higher* ($T_{\rm eq} \sim 2$–$3\times$ warmer), yielding a larger Jeans length and less refinement. This means the standard solver may *underestimate* the refinement needed in filaments and the circumgalactic medium.

2. Cooling overshoot near $T_{\rm eq}$.—The N-R solver's tendency to overshoot $T_{\rm eq}$ (Section 8.1.1) creates transient temperature errors that affect the *instantaneous* Jeans length even if the time-averaged temperature is correct. A cell that overshoots to $T < T_{\rm eq}$ will have a transiently small $\lambda_{\rm J}$, triggering refinement that may not be needed. Conversely, a cell that oscillates above $T_{\rm eq}$ will have a transiently large $\lambda_{\rm J}$, suppressing refinement that *is* needed. The exact integrator eliminates this noise entirely.

Table 4. Systematic differences between the RAMSES and Grackle/Townsend cooling implementations and their impact on the Jeans length.

Component	RAMSES	Grackle/Townsend	λ_J impact
Integration	N-R implicit (sub-stepping)	Exact Y-function (single step)	Removes overshoot noise
UV background	Courty (Cloudy 07)	Haardt & Madau (2012) (Cloudy 13)	T_{eq} shift $\rightarrow$ 40–80%
Metal cooling	CC07 CIE, linear Z scaling	3D non-LTE, density-dep. Z	Up to 10$\times$ at $10^{4.5}$ K
μ model	CIE, density-independent	Non-LTE, $\mu(n_H, Z, T)$	$\lvert\Delta\mu\rvert \sim 0.3$–0.6
Table dimension	2D: (n_H, T_2)	3D: (n_H, Z, T)	Metallicity resolution

3. Mean molecular weight.—The $T_2 = T/\mu$ representation amplifies errors in μ. With the RAMSES CIE model, μ is a function of T_2 alone (density-independent). The Grackle non-LTE μ depends on both n_H and T, with $|\Delta\mu| \sim 0.3$–0.6 in the transition zone ($T \sim 10^4$–10^5 K; bottom-right panels). Since $T = T_2\,\mu$, a 50% error in μ produces a 50% error in T and hence a $\sim 22\%$ error in λ_J, comparable to the factor-of-two ratio that separates adjacent refinement levels.

Table 4 summarises the systematic differences and their impact on the Jeans length.

9. DISCUSSION

9.1. *Physical Motivation*

The holdback mechanism was designed to prevent over-refinement from the mass criterion at early times. However, it inadvertently blocks the Jeans criterion, which has a different physical motivation: resolving the Jeans length prevents artificial fragmentation and ensures numerically converged star formation.

By allowing the Jeans criterion to operate ahead of the holdback release, we expect:

- Dense gas that will form stars is pre-refined before the holdback releases.
- When the holdback releases, the transition is smoother because the densest regions are already at the appropriate resolution.
- The SFR curve shows a gradual onset rather than an abrupt step.

9.2. *Interplay of the Three Parameters*

The bypass is controlled by three parameters that form a layered defence:

Layer 1: `jeans_bypass_holdback` *(on/off)*—Master switch. When off (default), the system is identical to standard RAMSES.

Layer 2: `d_jeans_thre` *(physics-based regulation)*—Restricts the Jeans check to dense gas where the polytropic floor operates. In this regime, $\lambda_J \propto \rho^{-1/6}$ varies slowly, and 1–2 extra refinement levels suffice to resolve it. Provides self-consistent, physics-driven cascade termination.

Layer 3: `n_jeans_bypass_levels` *(hard safety limit)*—Absolute cap on the number of extra levels. Prevents OOM regardless of physics configuration. Essential when `d_jeans_thre=0` (all cells checked).

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